



Sachin Tendulkar — Career Analytics

An innings-level statistical analysis of a legendary 24-year international career (1989–2013). Data sourced from ESPN Cricinfo, analyzed with Python, SQL Server, and Power BI.

DATA ANALYTICS PROJECT

782 INNINGS

34,357 RUNS

Career at a Glance

34,357

Total Runs

100

Centuries

164

Fifties

248

Highest Score

48.53

Batting Average

67.61

Strike Rate

782 innings · 74 not outs · Test, ODI & T20I · 15 Nov 1989 – 14 Nov 2013

Why This Dataset Matters

24 Years

Enables time-series and trend analysis across a full career

Multi-Format

Tests, ODIs, and T20Is offer rich comparative analysis

Multi-Geography

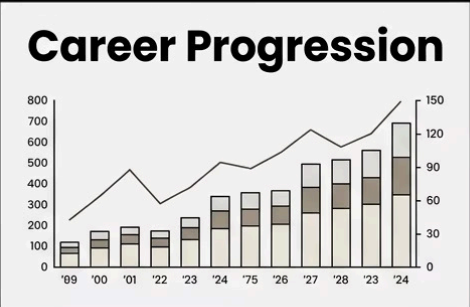
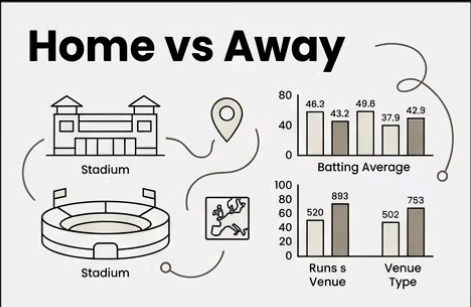
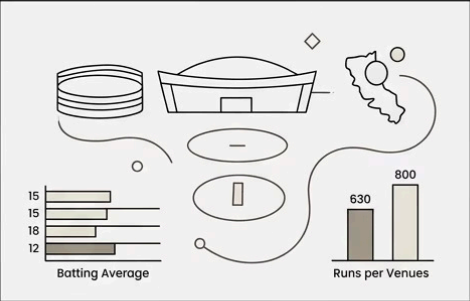
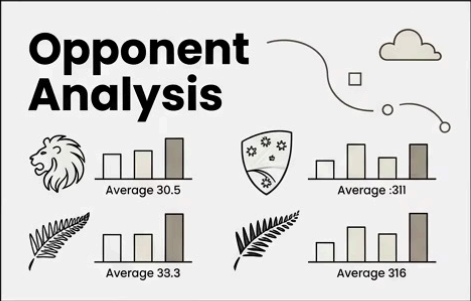
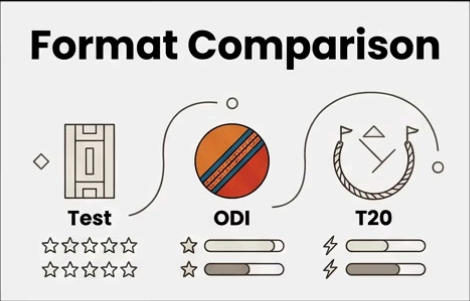
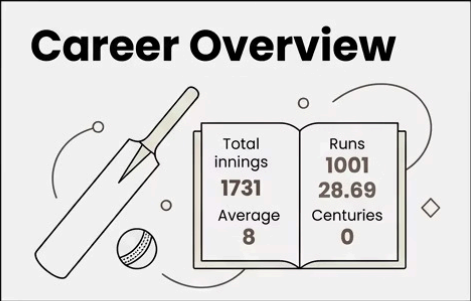
Home vs. away performance contrast across global venues

Milestone-Rich

100 centuries and 164 fifties – ideal for conversion and consistency analysis



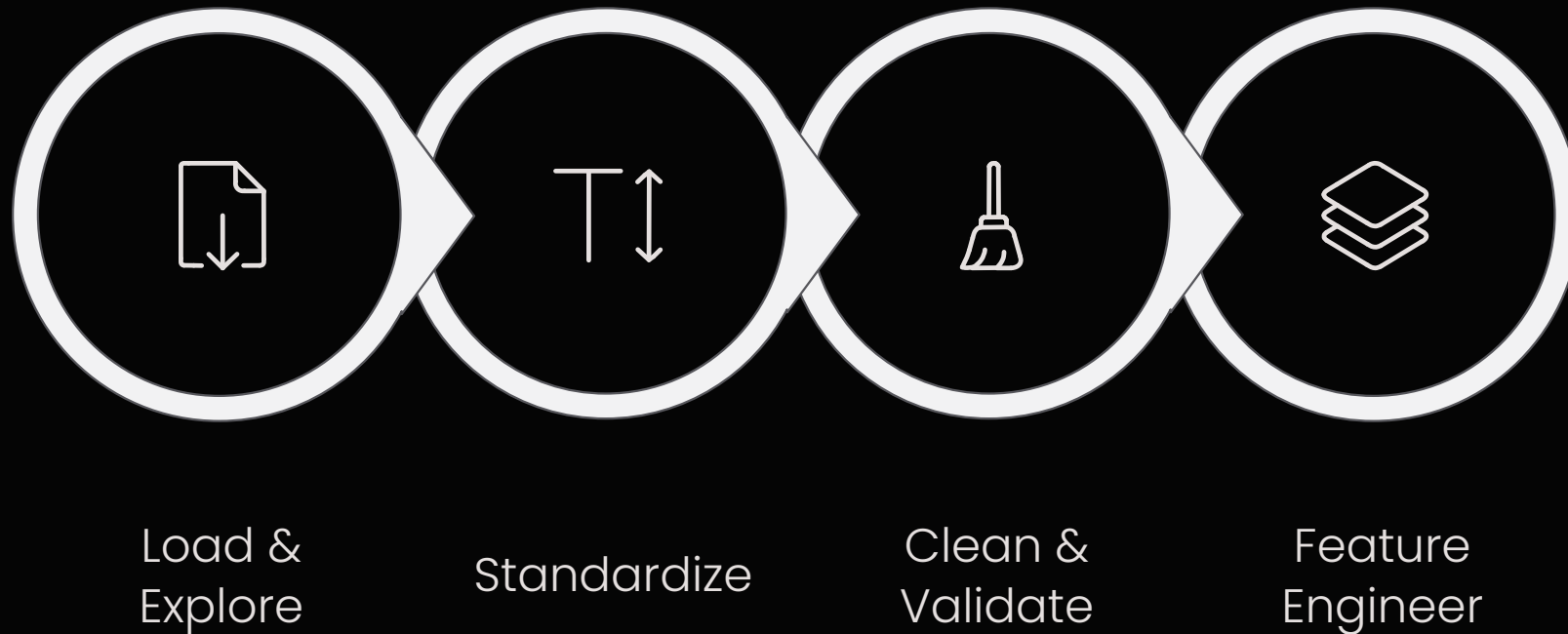
Business Questions



19 targeted business questions were answered across five analytical themes – from career summaries and format breakdowns to opponent analyses, dismissal patterns, and consistency metrics.

Data Pipeline: Python Cleaning

20 structured Python operations transformed raw ESPN Cricinfo data into a clean, analysis-ready dataset.



The cleaned dataset (782 rows) was exported to SQL Server via SQLAlchemy + pyodbc for structured querying and Power BI connectivity.

SQL Analysis — 19 Business Queries

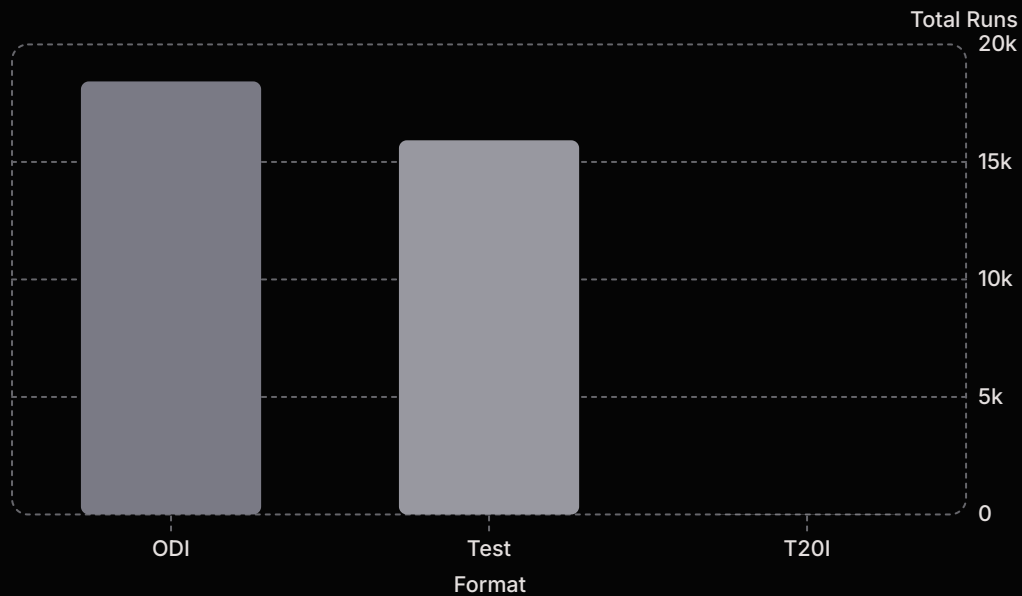
Key Techniques Used

- **GROUP BY** aggregations for career summaries
- **CASE WHEN** for scoring flags and conversions
- **Window functions** (SUM OVER) for cumulative runs
- **STDEV** for opponent consistency analysis

Sample Query — Career Summary

```
SELECT COUNT(*) AS innings,  
SUM(runs) AS total_runs,  
SUM(CASE WHEN runs >= 100 THEN 1  
ELSE 0 END) AS centuries  
FROM sachin_career;
```

Performance by Format & Opponent



Top Opponents by Runs (min 10 innings)

Australia

6,707 runs — his favorite opponent despite McGrath, Warne & Lee

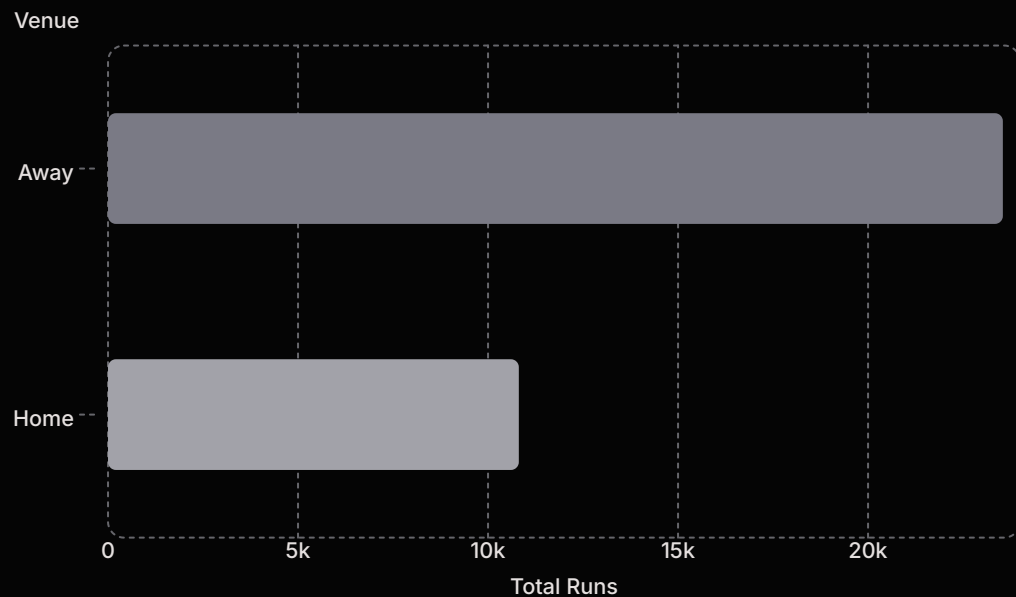
Sri Lanka

5,108 runs

England

3,990 runs

Home vs Away & Yearly Progression



Counter-Intuitive Finding

68.54% of Sachin's runs came away from home — proving he excelled on the toughest pitches worldwide, not just subcontinental turning tracks.

Peak Years

- **1998:** 2,541 runs — Desert Storm, Sharjah
- **2003:** 2,201 runs
- **1996:** 2,234 runs

Power BI Dashboard



Interactive Visuals

KPI cards, line charts, donut charts, cumulative area chart, and dismissal bar charts — all cross-filtering via slicers



Dynamic Slicers

Filters for Match Type, Opponent, Ground, Home/Away, and Year range enable deep exploration



Cricket-Themed Design

Dark background with gold KPI accents — "24 Years of Glory" branding with 1989–2013 span



Key Insights & Conclusion

100 Centuries

The only cricketer to reach 100 international centuries. Next best: Ricky Ponting (71).

Away Dominance

68.54% of runs scored away from home — class proven on the toughest pitches.

Conversion Rate

~37.88% of fifties converted to centuries — elite mental fortitude (typical range: 30–40%).

Format Adaptability

ODI strike rate grew over time; Test SR stayed steady (~50–55) — format intelligence.

 **The God of Cricket.** Consistent, adaptable, and relentless across formats, opponents, and geographies for 24 years. 1989 – 2013.